

Intent-Driven Machine Learning for Fake News Detection: A Referential Domain Adaptation Approach

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Abstract

Fake news negatively impacts business operations across various sectors, including healthcare, retail, and finance. One promising way to mitigate this problem is through automated content moderation using machine learning (ML). However, existing methodologies have struggled to identify fake news due to its inherent challenges—namely, class heterogeneity, concept drift, context dependency, and adversarial attacks. To address these challenges, we propose a novel strategy that focuses on the heterogeneous intentions behind fake news. Our approach builds on insights from prior social science research indicating that the intent behind creating fake news may differ from that behind disseminating it. This intent-driven variation influences the syntactic structures of fake news, leading to its heterogeneous nature. Accordingly, we argue that developing an ML model focused on the intent dimension of fake news can help overcome the issue of class heterogeneity, which often prevents effective model convergence. Moreover, since intent-driven linguistic patterns are generally less susceptible to temporal shifts and manipulative alterations, we expect this focus to enhance robustness against the other key challenges—concept drift, context dependency, and adversarial attacks. Grounded in a design science approach, we therefore propose a novel ML framework for fake news detection that centers on identifying and leveraging the intent behind fake news. Specifically, we first develop a domain adaptation model to infer the intent behind fake news, specifically targeting contexts where sufficient training data is unavailable—a common limitation in intent detection. Then, based on the intent detection model, we redefine fake news detection by shifting from binary (fake vs. true) to ternary classification (deceptive fake news vs. non-deceptive fake news vs. true news), thereby reducing the computational burden on ML models tasked with processing heterogeneous information within the fake news category. When evaluated against state-of-the-art baselines, our approach demonstrated significant improvements in both intent detection and fake news classification. Robustness checks further demonstrate that our method is less sensitive to distributional shifts in data and is more computationally efficient than baselines. Moreover, post-hoc analyses reveal the underlying mechanisms through which incorporating intent into fake news detection enhances model performance, addressing the key challenges associated with fake news.

Keywords

Fake News, Intent, Domain Adaptation, Deep Learning, Artificial Intelligence

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1 Introduction

The proliferation of fake news has a substantial adverse impact on the operational aspects of companies spanning various sectors, including healthcare, retail, and finance (Lee and Ram, 2024). Medical rumors disrupt the effectiveness of the healthcare supply chain by manipulating patients to forgo appropriate medical care and treatments (Liao et al., 2023). Likewise, the dissemination of fake news pertaining to a pandemic has unsettled consumer product demand, resulting in a cascading disruption of their supply chains (Xu et al., 2020). In finance,

fake news disseminated through social media channels encourages irrational investor behavior, leading to sudden surges in

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trading activity that pose risks to financial risk management (Zhang et al., 2022).

Recently, to address the issues posed by fake news within operations management, there has been a significant focus on the role of artificial intelligence (AI; Zhang et al., 2019). Building on recent advancements in AI, a multitude of studies have suggested that we harness its capabilities for the automated identification and moderation of fake news (Lee and Ram, 2024; Zhang et al., 2022). This aims to relieve the significant workload placed on human content moderators, responsible for handling the overwhelming volume of fake news content on a daily basis. Social media platforms like meta, previously reliant on human moderators, are gradually transitioning to AI-driven content moderation systems (Lee and Ram, 2024).

Against this backdrop, prior studies have explored various machine learning (ML) approaches, ranging from conventional ML¹ to pretraining-and-finetuning and prompt engineering (Guo et al., 2020; Huang et al., 2023; Jiang et al., 2023; Karadzhov et al., 2017; Wu et al., 2023). While these methods have achieved varying degrees of success, their effectiveness remains constrained by the inherent complexity and challenges of fake news detection (Lee and Ram, 2021; Wang et al., 2024).

Specifically, humans are particularly susceptible to fake news due to a range of cognitive limitations, including confirmation bias, the availability heuristic, the illusory truth effect, and negativity bias (Melchior and Oliveira, 2024). These psychological factors attract individuals from diverse backgrounds and with differing intentions to participate not only in creating but also in spreading fake news, thereby amplifying its heterogeneity and complicating algorithmic detection efforts. (Lee and Ram, 2024; Papanastasiou, 2020). For example, among fake news spreaders, some individuals may be aware of the malicious intent behind the content's creation but choose to feign belief because doing so aligns with certain personal or ideological benefits (*aka* preference falsification; Kuran, 1987). In contrast, others may genuinely believe the information to be true, lacking awareness of its deceptive origins. This variation in intent significantly complicates fake news detection, as both types of individuals may generate distinct versions of the same narrative based on their intentions—often reflected in different textual characteristics—and further disseminate them across social networks, amplifying the spread and transformation of misinformation (Lee and Ram, 2021). We call this *class heterogeneity*, since the distribution of data within the same fake news class is itself heterogeneous, which in turn leads to challenges for ML convergence. Additionally, certain evolving traits of fake news further complicate detection efforts, such as *concept drift*, where fake news adapts over time in response to changing environments; *context dependency*, where subtle shifts in framing increase perceived credibility; and *adversarial attacks*, in which creators deliberately alter content to improve its misleading effectiveness (Mohawesh et al., 2021).

To this end, we emphasize one aspect of fake news that has the potential to offer a solution to the aforementioned challenges but has received relatively little attention: the intent behind fake news. In prior literature, fake news, based on its intent, has been classified into misinformation and disinformation (Kumar and Shah, 2018). Misinformation refers to content that contains false information but is not created or shared with the intent to deceive, whereas disinformation is deliberately crafted to mislead others (Wang et al., 2024; Zhou et al., 2022). Unlike disinformation, misinformation typically arises from individuals who *spread* fake news rather than its creators, as they often consume and share the content *while believing it to be true* (Giglietto et al., 2019; Zhou et al., 2022). For example, while those who generate fake news linking the outbreak of monkeypox to coronavirus disease 2019 (COVID-19) vaccination may do so with deceptive intent (i.e., disinformation, henceforth referred to as *deceptive fake news*), those who receive, believe, and disseminate the news may spread it without deceptive intent, as they initially consider it to be true due to various cognitive biases (i.e., misinformation, henceforth referred to as *non-deceptive fake news*).

Depending on intent, individuals may exhibit distinct behaviors, which can manifest through variations in writing style (Fuller et al., 2009; Zhou et al., 2003). For instance, those with deceptive intent tend to use less consistent verb tenses, more complex sentence structures, and a higher frequency of hedging phrases that weaken their arguments compared to individuals without deceptive intent (Bachenko et al., 2008; Markowitz and Griffin, 2020). This distinction is particularly significant because the embedding vectors generated by language models (LMs) capture both the semantic meanings and syntactic structures of the text. Using these vectors as input for ML-based fake news detection models may lead to confusion, as the embeddings of data instances within the same class (i.e., fake news) can syntactically vary despite their semantic similarities. More critically, non-deceptive fake news may share syntactic characteristics with true news (because both non-deceptive fake news and true news have non-deceptive intent) while conveying different semantic content, leading to somewhat similar embedding vectors that can further complicate ML's information processing (Lee and Ram, 2021).

Figure 1² illustrates these issues in detail. It presents three news articles discussing the same topic but differing in content veracity and intention. Figure 1(a) and (b) together demonstrate how non-deceptive and deceptive fake news, despite conveying the same semantic information, exhibit distinct syntactic structures. Consistent with prior research on intent and writing styles, Figure 1(a) shows that non-deceptive fake news is written with consistent verb tense usage and a simpler sentence structure (Bachenko et al., 2008; Zhou et al., 2003). In contrast, deceptive fake news is crafted with a more complex sentence structure, passive voice, modal verbs, hedging expressions, and inconsistencies in verb tense usage (Bachenko et al., 2008; Fuller et al., 2009; Markowitz and Griffin, 2020). Even more notably, true news (i.e., Figure 1(c))

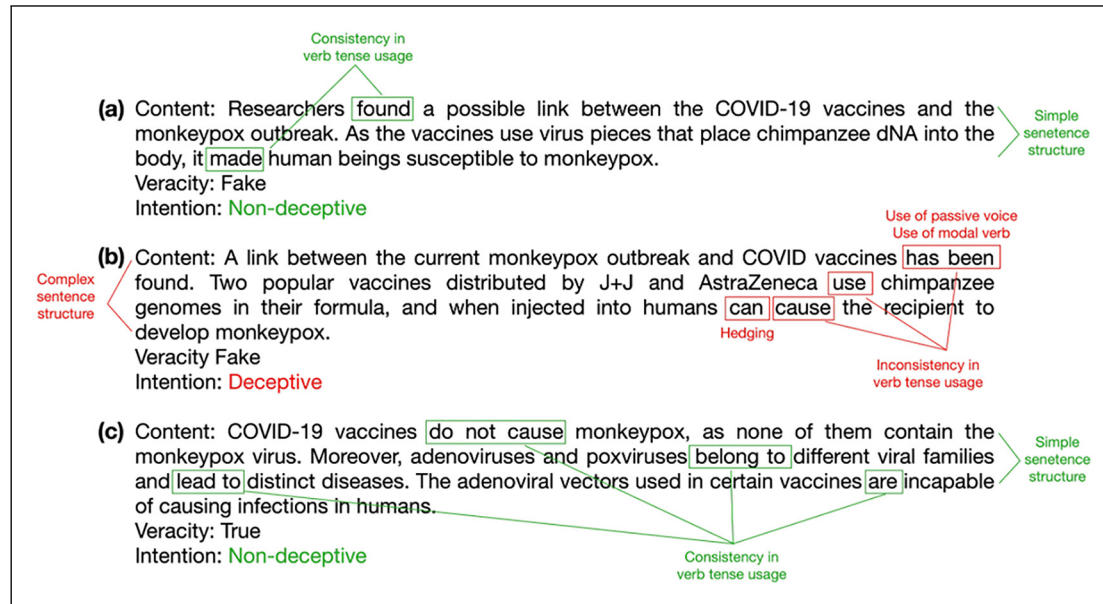


Figure 1. Examples of syntactic differences based on intent.

shares similar syntactic characteristics with non-deceptive fake news despite having entirely different semantic content, further highlighting the challenge for ML-based fake news detection models.

Incorporating the aspect of intent into the design of fake news detection models offers significant advantages for improving their performance from several perspectives. First, this approach can help address the issue of class heterogeneity. As shown in Figure 1, even within a single category of fake news, the distributions of syntactic structures vary considerably, contributing to class heterogeneity and hindering optimal ML convergence. Moreover, certain syntactic characteristics of fake news resemble those of true news—for example, non-deceptive fake news in Figure 1(a) compared to true news in Figure 1(c)—which further exacerbates the challenge of class heterogeneity. Incorporating the intent dimension into the design of fake news detection models can therefore provide an effective way to mitigate this problem.

Second, by training models to capture the intent dimension of text in addition to general semantic and syntactic information, they become better equipped to address concept drift—an issue that approaches relying solely on broad textual features cannot effectively manage. For example, the content of fake news related to COVID-19 evolved over time. In the early stages of the pandemic, malicious actors disseminated disinformation claiming that vaccination could be fatal. However, as public understanding of vaccines improved, the narrative shifted toward sowing doubt about potential side effects. This shift suggests that a detection model trained on fake news emphasizing the fatality of COVID-19 vaccines may fail to generalize to other streams of vaccine-related misinformation, as the semantics surrounding such content quickly evolve over time. In contrast, the underlying intent of fake news creators

remains constant, meaning that detection models leveraging intent are likely to be more resilient to concept drift (Horne et al., 2019).

Third, emphasizing the intent behind fake news can help alleviate the problem of context dependency in its detection. While the semantic meaning of text may shift with minor contextual changes—similar to the issue of concept drift—the writing styles that convey intent tend to remain more consistent and less susceptible to manipulation (Horne and Adali, 2017). For example, fake news asserting that people of color commit more crimes may appear semantically sound if contextual factors—such as the disproportionate policing in their residential areas—are not taken into account. However, as shown in Figure 1(b), while individuals deliberately craft fake news to omit such contextual cues and make it appear semantically coherent, their underlying intent often manifests in their writing and can be detected through methods designed to capture such intent.

Fourth, although some may attempt adversarial attacks by altering both content and syntactic structures of fake news to make it sound authentic, modifying syntaxes, especially those that reflect a writer's underlying intent, is typically more challenging for humans than adjusting semantics (Horne et al., 2019). Some may argue that LMs, with their ability to generate fluent and convincing text, could overcome this human limitation (Wu et al., 2024). However, as demonstrated in Figure 2, even advanced models struggle to entirely eliminate the traces of deceptive intent. In Figure 2, we present an example where GPT-4o attempts to modify the original text from Figure 1(b) to remove deceptive elements.³ While certain deceptive features, such as inconsistent verb usage and passive voice, have been reduced, others, including the use of hedging and modal verbs, still persist. This aligns with

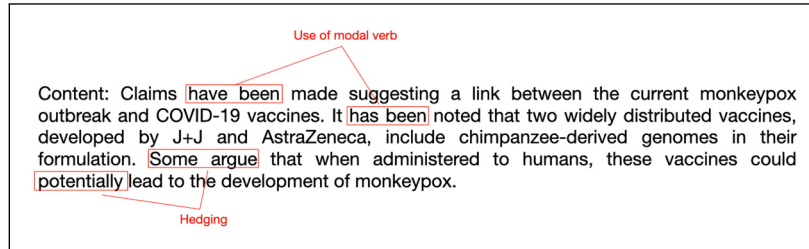


Figure 2. An example of syntactic embellishment by a large language model.

recent research examining the rhetorical, stylistic, and lexical features of language-model-generated text in comparison to human-generated content (Reinhart et al., 2025; Tang et al., 2024).

To that end, we propose a novel methodological framework for detecting fake news with a focus on intent, as outlined in Figure 3. Our study distinguishes itself from previous research in several keyways. First, unlike existing studies that primarily concentrate on how to craft input features or to design the ML architecture (e.g., different variations of transformers) for effective fake news detection, we rather propose transforming the output features of ML to guide it in more clearly understanding the heterogeneous nature of fake news. Specifically, as depicted in the speech bubble in Figure 3, we extend the conventional binary classification problem in fake news detection (i.e., true and fake news) into a ternary classification problem (i.e., true news, non-deceptive fake news, and deceptive fake news). Compared to prior research, which has mainly focused on providing intent-relevant features to ML (Lee and Ram, 2021; Wang et al., 2024), our approach more clearly reduces the complexity of ML by providing a direct link between input features and output classes.

Second, to implement this approach, we introduce a two-stage methodological framework that leverages domain adaptation (DA) to capture deceptive intent in text. An essential prerequisite for our ternary transformation of the fake news detection task is the identification of intent. Specifically, by detecting the presence of malicious intent behind fake news, we can further subclassify it, thereby reducing the complexity that ML models must handle. A major challenge in intent detection using ML is the lack of sufficient training data, unlike in other text classification tasks. Since the true intent of a writer is inherently unknowable, it is difficult for external annotators to reliably classify text into specific intent-based categories. In other words, generating training data for intent detection is challenging outside controlled lab settings. To address this challenge, we propose a two-stage approach. In the first stage, we develop a DA model in one domain using a limited but reliable set of ground-truth data for intent detection. In the second stage, we adapt this model to the fake news domain, which lacks the ground-truth data for intent detection, enabling it to classify deceptive and non-deceptive fake news more effectively.

Third, to tackle the well-known challenges of DA, such as negative transfer and catastrophic forgetting, and maximize the accuracy of intent detection, we introduce a novel approach called *referential DA*. Transferring knowledge from a source domain to a target domain presents significant challenges due to distributional inconsistencies between the two domains. Moreover, in the absence of ground-truth data in the target domain, the performance of the adapted model remains highly uncertain and cannot be validated. To reduce this uncertainty, we propose to a novel DA approach uniquely combining various techniques including contrastive learning, bootstrapping, and weight cloning. Specifically, we first identify a field of reference (henceforth, FoR) that shares the same distributional characteristics of source and target domains as intent detection, which is our field of interest (henceforth, FoI). Note that while the relationship between source and target domains in FoR is similar to that in FoI, the two fields involve different tasks. The purpose of introducing FoR is to build a model that predicts how DA performance will be influenced by distributional shifts between domains in one task with ground-truth labels and adjusts model parameters accordingly in another task without the labels. Subsequently, these prediction results are transferred, through the process called *weight cloning*, to FoI to mitigate negative transfer between domains with different data distributions but the same task, which could otherwise degrade the overall performance of our methodological framework.

The contribution of our study is multifold. First, from a methodological perspective, we extend the literature on fake news detection and DA. By untangling the complex nature of information generation and sharing—specifically, intent—which adds an additional layer of complexity to ML-based fake news detection, our new approach overcomes the challenges faced by prior studies while significantly improving model performance. To achieve this, we introduce a novel method for DA by proposing referential DA, which mitigates the negative effects associated with DA. Our approach also extends beyond the context of intent detection and can be applied to a broader range of tasks where the number of training data is extremely limited.

Second, from a theoretical standpoint, we contribute to the literature on false information and intent. While existing research on false information has long debated its intentionality, distinguishing between misinformation and

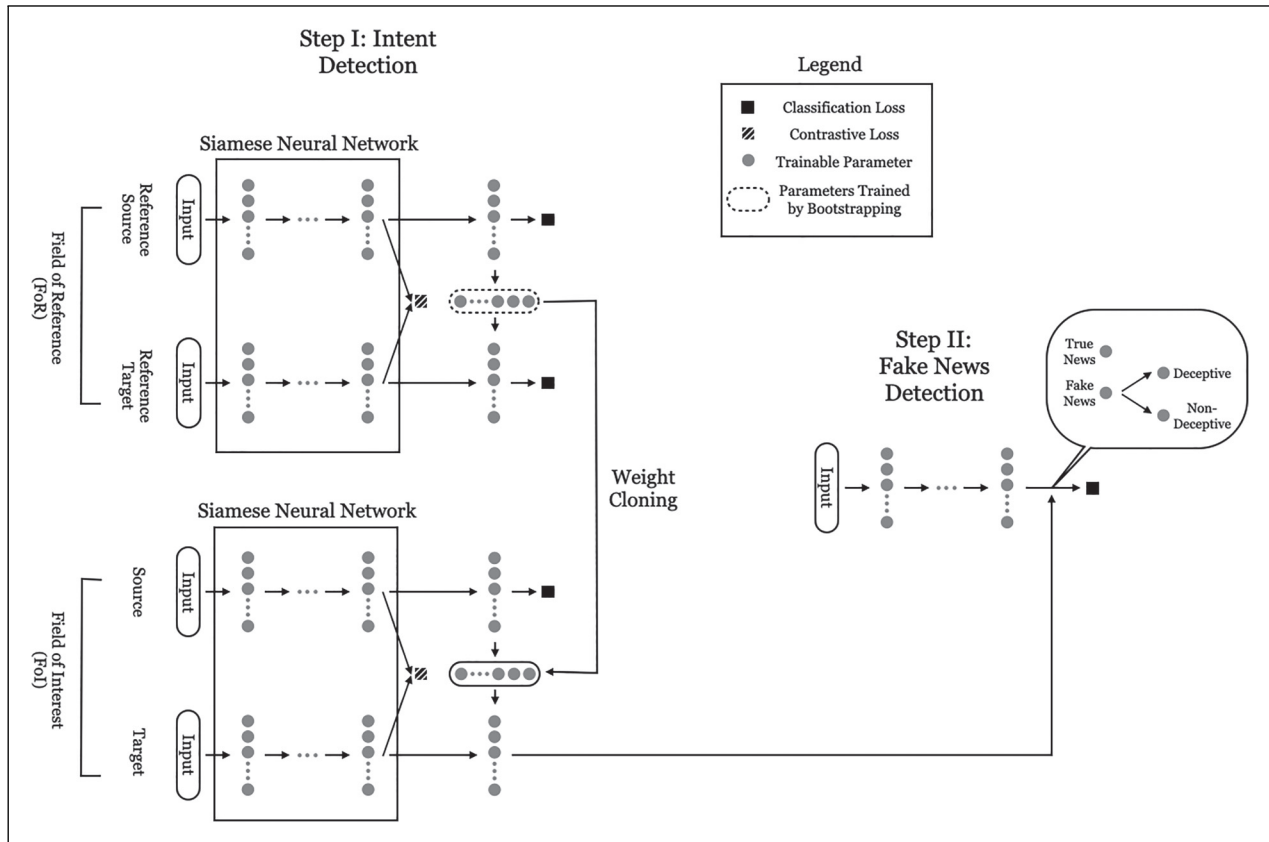


Figure 3. Proposed framework for fake news detection.

disinformation, few studies have taken an integrated approach that examines how intent plays a crucial role in shaping the syntactic structures of sentences. By focusing on this aspect, we demonstrate how true information, misinformation, and disinformation are interconnected through both syntactic and semantic relationships and offer a clear framework for addressing this complexity.

Third, we contribute to the literature on operations management in fake news. Despite being a disruptive factor for business operations, fake news has received relatively little attention. Although AI is a powerful tool for addressing the challenges of fake news, it encounters significant difficulties due to the inherent complexities of fake news—arising from individuals' susceptibility to misinformation, its viral nature, and additional factors such as concept drift, context dependency, and adversarial attacks. Our proposed method introduces a novel approach to developing fake news detection models that effectively mitigate the complexity.

Lastly, this study has significant practical implications, as our model can reduce the resource burden for businesses such as news publishers and social media platforms, which require substantial human resources for fake news moderation (CNBC, 2021). While automated methods like ours still require human oversight before implementation, they have the potential to considerably reduce the effort and costs currently

required for fake news moderation (CNBC, 2021; Lee and Ram, 2024).

2 Related Work

2.1 Fake News and Intent

Fake news comprises intentionally fabricated content designed to mimic the semantic and syntactic structures of genuine news articles (Moravec et al., 2020). Given that the primary intent behind fake news is to deliberately deceive readers into trusting its content, distinguishing fake news from genuine news articles poses a significant challenge, particularly for individuals lacking background information in the field (Shu et al., 2017).

To this end, numerous prior studies have focused on understanding the characteristics and motives of fake news producers (Moravec et al., 2020). These studies seek to answer fundamental questions such as their choice of topics, strategies employed to fabricate fake news that resemble genuine news, and how they embellish themselves as legitimate, alternative sources to traditional journalism (Robertson and Mourão, 2020). Based on the findings, different approaches—algorithmic filtering, educational campaigns, debunking initiatives—have been suggested to control and

regulate fake news producers and their contents (Lee and Ram, 2024).

However, fake news detection is far more complex than assumed in prior studies and requires going beyond simply identifying content produced by malicious actors. This is because the spread of fake news involves not only those who create it (hereafter referred to as *fake news producers*) but also those who disseminate it (hereafter referred to as *fake news spreaders*; Papanastasiou, 2020). While fake news producers consistently act with malicious intent when generating and distributing disinformation through social networks, fake news spreaders exhibit heterogeneous intentions—ranging from benign to malicious—which leads to significant textual variation even within the same fake news category, thereby complicating detection efforts (Lee and Ram, 2021; Zhou et al., 2022).

Although fake news producers and some spreaders are driven by self-serving motives such as personal promotion or the propagation of political ideology, other spreaders share misinformation for entirely different reasons, such as genuinely believing it to be true (Melchior and Oliveira, 2024). Indeed, prior studies have identified several cognitive mechanisms underlying spreaders' false belief in fake news, including desirability bias (the tendency to accept disinformation aligning with one's preferences), availability cascade (believing disinformation based on its popularity rather than accuracy), and the validity effect (increased trust in disinformation through repeated exposure; Lazer et al., 2018; Zhou et al., 2022).

When people have different intentions for spreading information, they exhibit distinct behavioral patterns (Fuller et al., 2009). These patterns can manifest in how they craft text to spread fake news, particularly in how they write about the news from both semantic and syntactic perspectives (Pennebaker, 2016). Previous studies have identified significant differences in writing styles between individuals with and without deceptive intentions, including variations in sentence length, pronoun usage, emotional expression, and more (Zhou et al., 2003). Since textual styles have been one of the most crucial information sources in automated fake news detection, capturing the intentions of fake news spreaders holds significant importance (Shu et al., 2017).

Specifically, variations in syntactic styles within the fake news class, caused by heterogeneous intentions, can hinder ML models from achieving optimal performance. This is because ML typically assumes that data samples within the same class share a consistent distribution; however, some fake news samples—such as those created with malicious intent versus those spread unintentionally—may exhibit divergent distributions (Lee and Ram, 2021).

Beyond its potential to enhance automated fake news detection, discerning the intent behind fake news has additional importance. That is, one of the common strategies to combat fake news on social media involves restricting accounts associated with the dissemination of fake news driven by

deceptive intent (Zhou et al., 2022). However, for this strategy to succeed, businesses must accurately discern whether posts on social media harbor deceptive intent or not. Without a means to identify the intent behind fake news, there is a risk of incorrectly targeting innocent parties (i.e., those with no lucrative motives), which could result in the breach of the First Amendment (Lee and Ram, 2021).

Moreover, in efforts to eliminate fake news from their platforms, businesses can achieve greater effectiveness by focusing on regulating the producers of fake news rather than its spreaders, thereby addressing the root cause of fake news propagation (Park et al., 2023). Identifying the intent behind fake news presents an effective approach to target fake news producers, contrasting with efforts that do not take into account deceptive intent (Zhou et al., 2022).

2.2 Automated Fake News Detection and Intent

To face the challenges posed by the overwhelming spread of online fake news, many researchers have suggested computational methods as a practical solution. Among different approaches such as information retrieval and knowledge graph, ML-based methods have been spotlighted in recent years (Shu et al., 2017). With a quality dataset for training, ML in general produces better results than other approaches such as rule-based methods, knowledge graphs, etc. Consequently, recent studies have focused on experimenting with various types of ML methods for fake news detection (Shu et al., 2017). These include long short-term memory (LSTM) neural networks, convolutional neural networks (CNNs), the attention mechanism, transformers, and graph neural networks (Guo et al., 2020; Karadzhov et al., 2017; Riedel et al., 2017; Shu et al., 2019; Yang et al., 2023)—which have employed various strategies, including word and graph embeddings, to transform raw data (e.g., text, reference network, etc.) into a desired type of input for ML models. However, as summarized in Table 1 (i.e., *Conventional ML*), these methods hold several limitations, including a strong dependence on the diversity and quality of training data, which restricts their effectiveness to the specific contexts represented within that data. Consequently, they remain vulnerable to major challenges in fake news detection, such as the heterogeneous nature of fake news, concept drift over time, shifting contextual dependencies, and semantic manipulation or embellishment introduced through adversarial attacks.

More recently, there has been a growing interest in using pretrained LMs without additional updates of their parameters, known as prompt learning, for automated fake news detection (*Prompt Learning* in Table 1; Gao et al., 2023; Huang et al., 2023; Jiang et al., 2022; Jiang et al., 2023; Wu et al., 2023). It is empirically known that the prompt learning approach often has an advantage over conventional ML in terms of using significantly fewer resources (Huang et al., 2023). Additionally, it reduces the gap between pretraining tasks and downstream tasks by transforming downstream tasks

Table 1. Summary of research gaps.

	Inclusion of the intent dimension of fake news	Dependence on ground-truth data	Resilience to the key challenges of fake news detection			
			Heterogeneity nature of fake news	Concept drift of fake news	Context dependency of fake news	Adversarial attacks
Conventional machine learning (ML) ²	No	High	Sensitive ⁵	Sensitive ⁶	Sensitive ⁶	Sensitive ⁷
Prompt learning ³	No	Low	Sensitive ⁵	Sensitive ⁶	Sensitive ⁶	Sensitive ⁷
Intent-augmented ML ⁴	Yes	High	Moderate ⁸	Moderate ⁸	Moderate ⁸	Moderate ⁸
Our study	Yes	Moderate ⁹	Robust ¹⁰	Robust ¹⁰	Robust ¹⁰	Robust ¹⁰

¹Intent-classification-focused ML approaches (Guo et al., 2023; Zhou et al., 2022) are excluded from the above summary, as their primary focus is not fake news detection.

²Guo et al., 2020; Karadzhov et al., 2017; Riedel et al., 2017; Shu et al., 2019; Yang et al., 2023.

³Gao et al., 2023; Huang et al., 2023; Jiang et al., 2022; Jiang et al., 2023; Wu et al., 2023.

⁴Conroy et al., 2015; Lee and Ram, 2021; Shu et al., 2019; Wang et al., 2024.

⁵All fake news is treated the same. Even prompt learning—which can be adapted to capture heterogeneity in fake news intent through simple prompt modifications—remains vulnerable to this issue, given that identifying others' intent from textual cues alone is challenging even for humans.

⁶Only the concepts and contexts presented in the training data can be understood.

⁷Sensitive to adversarial attacks involving the semantic manipulation of fake news.

⁸Implicitly manage the issue by incorporating the intent-based features into fake news detection.

⁹The weight cloning mechanism substantially reduces the amount of ground-truth data required.

¹⁰Robust to this issue by explicitly incorporating the intent dimension of fake news through our ternary-classification approach.

into masked language modeling tasks, thereby facilitating the use of knowledge embedded in pretrained LMs (Jiang et al., 2023). However, a notable downside of prompt learning is that it tends to underperform conventional ML approaches when there is sufficient data to update model parameters (Liu et al., 2023). In addition, although these methods utilize training data drawn from broader and more diverse sources than conventional ML, they still face issues of limited generalizability and suboptimal performance due to the inherent challenges posed by fake news.

Despite the various approaches discussed above, the intent behind the creation and spread of fake news has received relatively less attention. There are only few studies that have focused on intent as the primary interest in their fake news detection methods (*Intent-augmented ML* in Table 1). One stream of research has sought to incorporate intent-based features as input to ML to enhance their training process. For example, Wang et al. (2024) applied in-context learning with a large LM to infer the potential intent behind input text and used these inferred intent categories as augmented input for a fake news detection model. However, their approach relied on intent categories generated by the LM without considering the ground-truth intent behind fake news. They assumed that the model's reasoning about intent categories represented ground-truth values, which is not necessarily the case. Additionally, their method overlooked the possibility that fake news may have non-deceptive intent. Few other studies have leveraged deceptive syntactic cues to aid ML in detecting fake news (Conroy et al., 2015; Lee and Ram, 2021; Shu et al., 2019). However, these approaches also share similar limitations with convention ML, as they only implicitly consider the intent

underlying news generation and dissemination through manually engineered features (i.e., they do not directly distinguish deceptive fake news vs. non-deceptive fake news; Conroy et al., 2015; Lee and Ram, 2021; Shu et al., 2019). Moreover, they heavily rely on ground-truth data for intent identification (Lee and Ram, 2021).

Another stream of research has focused exclusively on identifying the intent behind fake news as its primary objective. Guo et al. (2023) proposed a method that integrates reinforcement learning and sequential learning to classify the intent behind fake news. Their model categorizes fake news into having intent types such as socialization, emotional venting, and rumor propagation. Similarly, Zhou et al. (2022) introduced a social-network-analysis approach to detect the intent of fake news. Unlike Guo et al.'s work, which examined various intent categories, they solely focused on the deceptive intent behind users spreading fake news on social media, aligning more closely with our study's perspective. However, these studies have not explored how identifying fake news intent influences ML's ability to differentiate between true and fake news. Additionally, their models rely on the assumption that their annotation process accurately captures the ground-truth intent behind the text. This assumption is problematic, as the content creators and annotators in their datasets are distinct, meaning annotators do not have direct knowledge of the creators' intent and can only infer it.

To address the above research gaps in incorporating deceptive intent into fake news detection, we propose a two-stage framework that expands the output variables of news classification by explicitly integrating intent. This design clearly differentiates between deceptive and non-deceptive forms of fake news, thereby enhancing the model's robustness to the

inherent heterogeneity of fake news. Moreover, by explicitly modeling intent, the approach strengthens resistance to concept drift, context dependency, and adversarial manipulation, since it relies more on intent-related textual cues and relatively less on semantic information which can be easily altered and manipulated. Furthermore, to overcome the limitation of scarce ground-truth data for deceptive intent, we introduce a novel referential DA method that combines weight cloning, bootstrapping, and contrastive learning to infer true intent values.

3 Method: Fake News Detector Based on Deceptive Intent

3.1 Problem Definition

As illustrated in Figure 3 (and further detailed as a pseudo code with relevant notations summarized in Online Appendix A), our model, which we call **Fake News Detector based on Intent (FENDI)**, follows a two-stage approach, addressing two key problems: Intent detection (FENDI-1) and fake news detection (FENDI-2). As stated above, while the primary objective is to detect fake news, solving the intent detection problem is a key step to enhance the effectiveness of the overall model.

The intent detection problem, which serves as the initial task, can be formulated as finding an optimal function, $f_{int_target} : D_t \rightarrow D_{tY}$, where D_t represents an input dataset, and $D_{tY} \in \{0, 1\}$, with 0 and 1 denoting non-deceptive and deceptive intent, respectively. Note that the ground truth values for D_{tY} do not exist as the intent of the writers of D_t is unknown and must be inferred through a DA process, which will be detailed in the subsequent section.

The problem of fake news detection can be defined as finding an optimal function, $f_{news} : D_X \rightarrow D_Y$, where D_X indicates an input dataset containing potential fake news articles, and $D_Y = \{y'_i | t \in D_{tY}, i = \{1, 2, \dots, R\}, y'_i \in \{0, 1\}\}$, with R denoting the cardinality of the dataset and the range of y'_i (i.e., 0 and 1) corresponding to true and fake news, respectively.

3.2 FENDI-1: Referential Domain Adaptation for Intent Detection

As stated earlier, the primary challenge in intent detection is the limited availability of data. In only a few specific domains, intent detection datasets with ground-truth labels exist, which can be utilized for ML training. However, in the fake news domain, such data is unavailable. Some prior studies have attempted to address this gap by using human-annotated data, based on the overly strong assumption that annotators—who are not the original news creators—can accurately infer the intent behind the news solely by reading its content (Zhou et al., 2022). However, this assumption does not necessarily hold true in most, if not all, cases. To overcome this limitation, FENDI-1 has been designed to train an ML model capable of inferring intent behind text without relying on ground-truth labels by introducing a novel DA process.

The core component of FENDI-1 is the set of parameters for weight cloning, which learns how classification results for a specific task shift between a source domain and a target domain (represented by the parameters highlighted with a dashed oval in Figure 3 and denoted as f_{weight} in Figure A1). Specifically, given two domains with different data distributions but the same classification task, this component learns how the model's parameter weights should adjust from those of the source domain to those of the target domain to transfer knowledge for ensuring optimal task performance. These parameters do not capture general distributional differences between domains, that is, $distance(P(X_S), P(X_T))$. Instead, they reflect distributional differences between domains with respect to the classification task, that is, $distance(P(Y|X_S), P(Y|X_T))$. In other words, their purpose is not to indicate how two datasets differ in general but to guide how the target data should be utilized differently to achieve consistent classification results, given its differences from the source data.

To train these parameters, we employ three interrelated strategies: (i) Identifying proper FoR, (ii) contrastive learning, and (iii) learning through bootstrapping. A proper setup of FoR is essential, as effective knowledge transfer to FoI relies on the comparability of the source-target-task triplets in FoR and FoI. This alignment ensures that the learned representations are relevant and transferable. Furthermore, our incorporation of contrastive loss alongside classification loss plays a critical role in guiding the weight cloning component to capture $distance(P(Y|X_S), P(Y|X_T))$, not $distance(P(X_S), P(X_T))$. Lastly, the inclusion of bootstrapping serves to enhance the stability and convergence of the training process, helping the model reach at least a local optimum. We elaborate on the details of each component below.

First, we set up FoR, a set of source and target domains and their shared task. FoR should share the same source-target relationship as FoI but involves a different classification task. For instance, in FoI, FENDI-1 designates business review data as the source domain and news article data as the target domain, and intent detection as their shared task. FoR must maintain this same source-target domain relationship (i.e., business reviews and news articles), though it does not necessarily need to involve the task of intent detection. The only required condition for the task in FoR is that it must contain data annotated with ground-truth values for both source and target domains so that the parameters for weight cloning can be properly trained. As will be elaborated on in a later section, in this study, we define FoR using the task of sentiment detection while preserving the same source-target domain structure of business reviews and news articles.

Then, we construct a model to train the parameters for weight cloning. Specifically, for FoR, we design a model based on a Siamese neural network, where the model parameters are shared across the training processes of source and target domains (f_{ref} in Figure A1). On top of the Siamese neural network, we incorporate two additional feed-forward neural networks, each dedicated to classifying data from either the

source or target domain (f_{ref_source} or f_{ref_target} in Figure A1). These parameters are optimized through two types of loss functions: the contrastive loss (L_{rc} in Figure A1) and the classification losses for the source and target domains (L_{rsc} and L_{rtc} in Figure A1).

This dual-loss approach is grounded in the assumption that discrepancies between source and target domain data arise from two distinct sources: (i) General distributional differences unrelated to any specific task, that is, $distance(P(X_S), P(X_T))$, and (ii) task-specific differences relevant to the classification objective, that is, $distance(P(Y|X_S), P(Y|X_T))$, as denoted above.

The objective of weight cloning is to isolate and identify differences in model parameters that encapsulate task-specific data distributions of the source and target domains in FoR. These identified differences are then used along with the source-domain classification parameters in FoI to infer the corresponding parameters for target-domain classification in FoI.

To achieve this, we decompose the model architecture into two key components: The Siamese neural network layers, which are responsible for domain-general encoding of text samples from both domain datasets; and the task-specific classification layers, which learn the mapping between these encodings and the given task labels. The inclusion of contrastive loss is critical for enforcing this architectural separation. Without contrastive loss, backpropagation of the classification loss alone would affect both the Siamese neural network layers and the classification layers. This coupling would obscure the distinct roles of each component, as both sets of layers would jointly encode domain and task-specific differences—leading to blurred functional boundaries.

We propose our contrastive loss defined as follows:

$$L_{rc} = \alpha \left(\frac{1}{d(f_{ref}(x_n^{rs}), f_{ref}(x_m^{rs}))} + d(f_{ref}(x_n^{rs}), f_{ref}(x_m^{rs})) + d(f_{ref}(x_n^{rt}), f_{ref}(x_m^{rt})) \right), \quad (1)$$

where $\frac{1}{d(f_{ref}(x_n^{rs}), f_{ref}(x_m^{rs}))}$ measures the inverse of the distance between the embeddings of input data for the source domain and target domains; $d(f_{ref}(x_n^{rs}), f_{ref}(x_m^{rs}))$ estimates the within-domain distance of the source domain with random sampling of a data instance (x_m^{rs}); $d(f_{ref}(x_n^{rs}), f_{ref}(x_m^{rs}))$ denotes the within-domain distance of the target domain with random sampling of a data instance (x_m^{rt}); and α is a damping factor. Through this process, the model learns the overall distributional difference between the source and target domain data (i.e., $distance(P(X_S), P(X_T))$).

Then, we have two classification losses that are defined as follows:

$$L_{rsc} = \ell(f_{ref_source}(f_{ref}(x_n^{rs})), y_n^{rs}), \quad (2)$$

$$L_{rtc} = \ell(f_{ref_target}(f_{ref}(x_n^{rt})), y_n^{rt}), \quad (3)$$

These losses are designed for learning the classification task in the source and target domains, respectively, ensuring that the model distinguishes how the task should be handled differently in each domain while adapting to general distributional differences, as formulated in Equation 1.

The difference between the weights of f_{ref_source} and f_{ref_target} corresponds to how the same classification task can yield distinct results due the domain shift (i.e., $distance(P(Y|X_S), P(Y|X_T))$). Therefore, if we can learn the necessary weight adjustments to transform f_{ref_source} into f_{ref_target} , it becomes possible to accurately estimate f_{int_target} without requiring the ground-truth labels. In other words, by understanding how the classification layer weights adapt across domains for one task, we can infer how to structure the classification layer for the target domain in another task, even without ground-truth labels. This is achieved by extrapolating the weights of the classification layer in the target domain from those of the source domain, where ground-truth labels are available. To implement this, we develop a feed-forward neural network model (f_{weight} in Figure A1) that predicts the target-domain classification layer weights (w_{target}) based on the source-domain classification layer weights (w_{source}).

To train f_{weight} , we generate data within FoR using a bootstrapping approach. Specifically, we randomly sample subsets of data from both source and target domains and train the model components for FoR, including f_{ref} , f_{ref_source} , and f_{ref_target} . This process is repeated N times, resulting in N pairs of w_{ref_source} and w_{ref_target} . These weight pairs serve as training data for developing f_{weight} .

Once f_{weight} is developed, we apply the same procedure to our FoI—intent detection. However, in this case, we can only train f_{int} and f_{int_source} because the ground-truth labels for the target domain are unavailable, preventing direct training of f_{int_target} . To overcome this, we apply “weight-cloning” by using f_{weight} to infer f_{int_target} based on w_{int_source} , ensuring that the model adapts effectively despite the lack of labeled data in the target domain.

3.3 FENDI-2: Ternary Fake News Classification

During the training of FENDI-2, our fake news detection model, we reformulate the original binary classification task—distinguishing fake news from true news—into a ternary classification task. Specifically, within the training dataset, for new samples annotated as fake news, we apply f_{int_target} of FENDI-1 to further classify them into either deceptive or non-deceptive fake news. As a result, the fake news detection task transitions from a binary classification problem (i.e., true news vs. fake news) to a ternary classification problem (i.e., true news vs. non-deceptive fake news vs. deceptive fake news). This approach helps mitigate the complexity faced by ML models, due to the heterogeneous nature of fake news in terms of intent and the overlapping characteristics between true news and non-deceptive fake news.

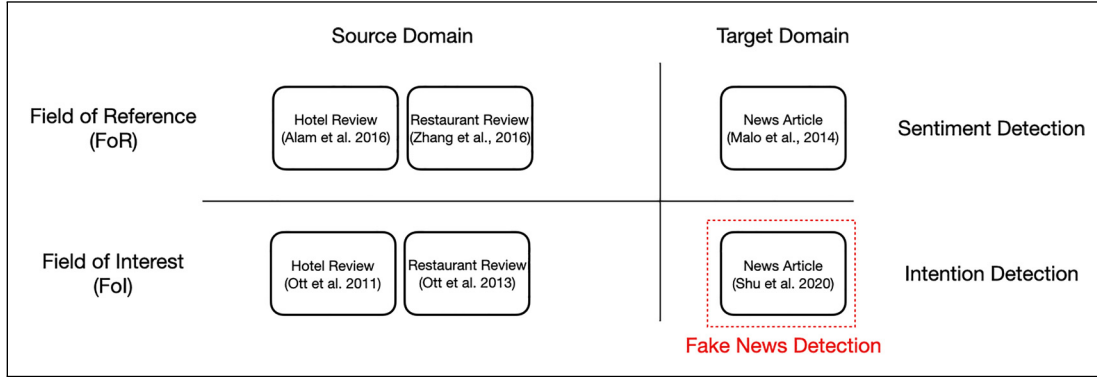


Figure 4. Summary of datasets.

Subsequently, we train f_{news} to learn the distinction between the three classes of news. Specifically, it consists of a base LM with a finetuning layer on top, which can be represented as follows:

$$f_{news}(x_i) = \sigma(W_f \cdot f(x_i) + b_f), \quad (4)$$

where W_f and b_f are the weights and biases of the finetuning layer, $f(x)$ is a pretrained LM, and σ is a non-linear activation function such as softmax. During inference, we merge the two subclasses of fake news (i.e., deceptive and non-deceptive fake news) into a single class (i.e., fake news) since the ultimate goal of our method is to differentiate between true news and fake news in a binary fashion.

4 Experiment

4.1 Data

We utilized multiple datasets to develop and evaluate our method, as illustrated in Figure 4. First, in FoR, we included two groups of datasets that share the same task of sentiment detection but originate from different domains—business reviews (i.e., source domain) and news articles (i.e., target domain).

For sentiment detection in business reviews (i.e., source domain), we used two publicly available datasets provided by Alam et al. (2016) and Zhang et al. (2015). The dataset from Alam et al. (2016) consists of 20,491 hotel reviews collected from Tripadvisor.com, including review text and corresponding ratings on a scale from 1 to 5. Since the rating distribution was skewed toward higher scores, with most ratings being 4 or 5, we considered only 5-star reviews as positive sentiment and combined 1- and 2-star reviews as negative sentiment. The resulting dataset contained 9054 positive reviews and 3214 negative reviews. In contrast, Zhang et al. (2015) provided an equally distributed dataset of 598,000 positive and negative restaurant reviews collected from Yelp.com.

For sentiment detection in news articles (i.e., target domain), we used the dataset from Malo et al. (2014), which contains 4837 human-annotated news articles classified into

positive, neutral, and negative sentiment categories. Specifically, 1363 articles were labeled as positive, 604 as negative, and the remaining as neutral. Since the FoR task involves binary classification, we excluded the neutral data and retained only the positive and negative labeled instances.

In FoI, we used two datasets as source domain data (Ott et al., 2011; Ott et al., 2013) and one dataset as target domain data (Shu et al., 2020). The datasets from Ott et al. (2011) and Ott et al. (2013) contain ground-truth labels distinguishing deceptive and non-deceptive intent for reviews in two business domains—hotels and restaurants. The non-deceptive reviews were collected from online platforms such as Yelp and Hotels.com, while corresponding deceptive reviews were crafted using MTurk (see references for details). This process resulted in 1600 (800 deceptive and 800 non-deceptive) hotel reviews and 400 (200 deceptive and 200 non-deceptive) restaurant reviews.

The target data in FoI, which also serves as the fake news detection dataset, consists of a combination of two datasets collected from PolitiFact.com and GossipCop.com. These platforms aggregate potential fake news in the political and entertainment domains from various social media sources (e.g., Facebook, Instagram, X) and provide expert evaluations on their veracity with fine-grained labels (Wang, 2017). While expert-based fact-checking sites do not scale as efficiently as other sources, they provide more accurate labels than crowd-sourced fact-checking datasets or computationally generated labels (Shu et al., 2017). Using the software provided by Shu et al. (2020), we collected 458 true and 351 fake news articles from PolitiFact.com, along with 4307 true and 4307 fake news articles from GossipCop.com.

4.2 Development and Evaluation of FENDI-1

As base models for FENDI-1, we selected relatively small pretrained LMs, specifically those with fewer than 100 M parameters, to ensure computational efficiency when developing the bootstrapping-based model in FoR. Specifically, we evaluated ALBERT (base; 11 M parameters), DistilBERT (base; 66 M parameters), and DistilGPT2 (82 M parameters).

For each base model, we explored various hyperparameter configurations, including the number of bootstrapping iterations, the number of epochs per iteration in the bootstrapping process, learning rate, batch size, and damping factor for regulating the influence of the contrastive learning component. A summary of the tested hyperparameter options is presented in Online Appendix B. In addition, further details of our model architecture are provided in Online Appendix C.

To assess the robustness of our proposed model to domain shift, we employed two evaluation strategies. First, as a preliminary test, we used hotel and restaurant reviews, alternating their roles as source and target domains across two iterations. Specifically, in the first iteration, hotel reviews served as the source domain, while restaurant reviews were assigned as the target domain. In the second iteration, their roles were reversed, with restaurant reviews becoming the source domain and hotel reviews acting as the target domain. Although the domain shift from hotel reviews to restaurant reviews (and vice versa) does not precisely reflect the domain shift encountered when adapting a model developed with business reviews to news articles, that is, our main focus of DA, we conducted this preliminary test because both datasets contain ground-truth labels so that we can measure how our strategy of weight cloning works effectively. This allowed us to preliminarily evaluate the model's ability to generalize across different but related domains before applying it to news articles, where ground-truth labels are unavailable.

To evaluate the robustness of our model under the domain shift of focus—from review data to news article data—we conducted an additional evaluation. Given the challenge of lacking ground-truth labels for news article data, we addressed this issue by collecting additional data on deceptive and non-deceptive fake news through Prolific.com. Specifically, we first selected ten fake news articles from DW.com, which provides detailed evidence explaining why these articles contain false information and used them as seed articles for data collection. Participants on Prolific.com were randomly assigned with one of the seed articles and asked whether they believed the content to be true or false, followed by completing a task corresponding to their stated belief: (i) If they *believed* the content, they were simply asked to paraphrase it, thereby generating non-deceptive fake news; (ii) if they *did not believe* the content, they were instructed to paraphrase it in a way that made it appear convincing to others, thus creating deceptive fake news.

In both cases, we asked them not to change the argument (i.e., semantics) made in the article. In addition, to minimize the influence of the original text on participants' paraphrasing process, we prevented them from returning to the survey page where the original article appeared while completing the task. As a result, we collected a total of 298 data instances, consisting of 146 non-deceptive and 152 deceptive cases. Using this dataset as the target data in FoI (i.e., test dataset), we trained FENDI-1 with a combination of hotel and restaurant review datasets as the source data to evaluate its ability to infer the intent behind fake news.

To evaluate model performance, we use the following three metrics: (i) Precision (i.e., the ratio of correctly classified deceptive reviews to predicted deceptive reviews) = $\frac{TP}{TP+FP}$; (ii) recall (i.e., the ratio of correctly classified deceptive reviews to actual deceptive reviews) = $\frac{TP}{TP+FN}$; and (iii) F1-score (i.e., a harmonic mean of the precision and recall rates) = $2 \cdot \frac{\text{precision} \cdot \text{recall}}{\text{precision} + \text{recall}}$.

As baseline models, we adopt state-of-the-art approaches from different ML schemes: *Conventional ML*, *pretrain-and-finetune*, *in-context learning*, and *soft prompting*. First, conventional ML leverages customized neural-network models such as LSTM and CNN to learn task-specific knowledge from data. The advantage of this approach is that it enables models to learn knowledge solely relevant to a given task without interference from learning other tasks. However, the downside is that it cannot distill knowledge to similar or general tasks. Among various deep learning methods, we include LSTM and CNN as baselines in this study, adopting the configuration from Wang (2017).

Second, unlike conventional ML, the pretrain-and-finetune approach harnesses general linguistic knowledge learned by a large, pretrained LM by adding a finetuning layer on top (Devlin et al., 2019). In this study, we adopt BERT (large; 340 M) and DistilBERT (base; 66 M), among many variations of transformer-based models, finetuning them with a feed-forward neural-network layer on top (768-dimension). The rationale for selecting BERT and DistilBERT is based on their distinct characteristics. BERT contains more knowledge due to its large number of parameters but requires more training data for effective finetuning (Devlin et al., 2019). DistilBERT, on the other hand, may be more appropriate in contexts where available data is limited since it has a shallower neural-network architecture.

Third, in-context prompt learning with few-shot examples shown to a pre-trained LM has become the most parameter-efficient approach for solving downstream tasks (Lee and Ram, 2025; Liu et al., 2023). This approach does not require any addition or update of parameters to the existing model, allowing it to solve any downstream tasks without computational bottlenecks. We use GPT-4o with 6 examples as an input prompt (3 true news articles and 3 fake news articles) for our classification task.

Lastly, we test cutting-edge soft prompting methods for DA, including those proposed by Lester et al. (2021) and Wang et al. (2022), as additional baselines. To implement Wang et al.'s method, we set hotel and restaurant reviews as separate domains with distinct tunable prompts. During inference, we applied *k*-means clustering to determine the closer domain for each data instance (i.e., news article) and select the appropriate prompt accordingly to ensure optimal classification.

We summarize our results in Table 2, where we find several notable findings. First, DA in intent detection is an extremely challenging task, and general models not specifically designed for it achieve only limited success. For example, while large

Table 2. Summary of results (intent detection).

Model		Domain adaptation direction								
		Hotel > Restaurant			Restaurant > Hotel			Review > News		
		Precision	Recall	F1	Precision	Recall	F1	Precision	Recall	F1
Conventional ML	LSTM (Wang, 2017)	85.28	70.00	76.89	54.99	93.75	69.32	74.07	52.63	61.54
	CNN (Wang, 2017)	86.18	68.90	76.58	53.29	95.35	68.37	69.57	42.11	52.46
Pretrain- and-Finetune	BERT	88.06	66.04	75.48	51.59	99.50	67.95	71.72	68.42	70.03
	DistilBERT	87.71	75.83	81.34	62.29	65.87	64.03	65.62	55.26	60.00
In-Context Learning	GPT-4o	50.00	12.50	20.00	56.82	25.50	35.20	53.33	42.11	47.06
Soft prompting	Lester et al. (2021)	78.87	84.00	81.35	55.10	71.50	62.24	66.23	67.11	66.67
	Wang et al. (2022)	80.18	85.00	82.52	56.85	67.87	61.87	69.14	73.68	71.34
FENDI-1 ^{1,2}		91.53	81.00	85.94	75.67	66.50	70.79	76.54	81.58	78.98

ML = machine learning; LSTM = long short-term memory; CNN = convolutional neural network.

¹Among the three base models for applying our method (i.e., ALBERT, DistilBERT, and DistilGPT2), we report only the one that perform well with computational efficiency, ALBERT, for brevity.

²The mean squared error of the weight cloning layer was 0.0018.

³The shaded areas indicate the highest scores for the corresponding columns.

LMs like GPTs and Llamas demonstrate versatility in various natural language processing tasks, the exceptionally poor performance of GPT-4o in our setting (F1-scores of 20.00, 35.20, and 47.06 for hotel-to-restaurant, restaurant-to-hotel, and review-to-news DA, respectively) indicates their unsuitability without further fine-tuning with additional training data. On the other hand, FENDI-1, designed to address the data scarcity issue in intent detection through an effective DA approach, significantly enhanced baseline performance, achieving F1-scores of 85.94, 70.79, and 78.98⁴ for hotel-to-restaurant, restaurant-to-hotel, and review-to-news DA tasks, respectively.

Second, while additional parameter updates with a small dataset improved ML model performance, the enhancement was limited without an effective DA technique. In Table 2, although models that underwent additional training (e.g., approaches other than in-context learning) showed significant improvements in intent detection, their influence was limited by dataset availability. For instance, in the hotel-to-restaurant setting, where a modest amount of training data was available (1600 instances), additional training proved effective in enhancing performance. Notably, models requiring updates to a relatively small number of parameters, such as DistilBERT and soft-prompting-based approaches, benefited the most, achieving F1-scores slightly above 80 points. Conversely, in the restaurant-to-hotel setting, where training data were extremely limited (400 instances), additional training of models failed to improve performance. Even the models that required updating only a small number of parameters did not benefit much from additional training, producing F1-scores less than 65 points. In addition, models that required training from scratch or involved substantial parameter updates faced significant difficulty in learning the intent detection task, as evidenced by the performance of LSTM, CNN, and BERT,

largely misclassifying data instances by labeling most as having deceptive intent. Nonetheless, in both cases, FENDI-1 consistently enhanced performance, highlighting its robustness to data scarcity.

Third, even with a larger dataset, significant domain shifts make intent detection highly challenging. In the review-to-news context, we combined hotel and restaurant review datasets to create a larger training set. However, due to the greater discrepancy between the review and news domains compared to that between hotel and restaurant reviews, baseline models performed poorly, achieving at best around 70 points in F1-score. In contrast, FENDI-1 effectively mitigated the domain shift issue by introducing the referential DA approach, achieving an improved F1-score of 78.98.

Overall, the results highlight the versatility of our approach in capturing both similarities and differences across domains in intent detection. Grounded in these findings, we decided to use FENDI-1 to further categorize fake news labels into deceptive and non-deceptive fake news for the next step.

4.3 Development and Evaluation of FENDI-2

Using FENDI-1, we classify fake news in the fake news dataset into two subclasses: Deceptive and non-deceptive fake news. Based on these new labels, we develop a deep learning model (i.e., FENDI-2) for a ternary classification task of identifying true, deceptive fake, and non-deceptive fake news. The ternary outcome of this model will be converted into binary classes (i.e., true and fake news) by merging deceptive and non-deceptive fake news into a single class after classification. Since our approach is method-agnostic, it can be tested using multiple pretrained LMs as base models. As our final pairing choice, ALBERT (base; 11 M parameters) has been selected balancing both performance and computational efficiency. As baselines, we incorporate a similar set of methods used to

benchmark FENDI-1, along with a few additional models from prior literature specifically designed for fake news detection.

For the conventional approaches, we employ LSTM, CNN, and the method proposed by Liu and Wu (2020). Similar to FENDI-1, the model architectures of LSTM and CNN have been adopted from Wang (2017). Additionally, Liu and Wu (2020) proposed a CNN-based method to detect fake news on social media. Since their method requires personal information such as the number of followers and the number of tweets, we have omitted the components of their model architecture that depend on such data.

For the pretrain-and-finetune approach, we utilized BERT and DistilBERT, guided by the same rationale applied in choosing the base models for FENDI-1. The configurations for BERT and DistilBERT were maintained consistently with those employed in FENDI-1. Moreover, we evaluated the approach suggested by Kaliyar et al. (2021), which integrates BERT embeddings with several layers of CNN.

For prompt learning, we incorporated several state-of-the-art techniques that utilized similar datasets as ours to evaluate their performance. Specifically, Peng et al. (2023) proposed in-context prompt learning combined with data augmentation for detecting COVID-19-related misinformation. As the data augmentation aspect is not relevant to our context, we have solely adopted their approach to prompt formatting for in-context learning. Hu et al. (2024) proposed a hybrid approach combining prompt learning and fine-tuning, where rationales generated by prompt learning are selectively utilized in the fine-tuning process. We also examined the prefix-tuning method employed in FENDI-1 (Lester et al., 2021). It is important to note that the soft prompting techniques used in FENDI-1, such as those by Wang et al. (2022), were excluded from our consideration as their model architectures were tailored for acquiring common knowledge across various domains, whereas our fake news detection task is confined to a single, specific domain. Note that, for all methods, we have explored the sets of hyperparameters as suggested in Online Appendix B. We evaluate each model using the five-fold cross-validation with the same metrics that we have used for the evaluation of FENDI-1.

In Table 3, we summarize the results on both PolitiFact and GossipCop. The results highlight a couple of important points worth noting. First, FENDI-2 showed the strongest performance on both PolitiFact and GossipCop datasets. Specifically, it improved the performance of the other models by 8.48–23.14 in F1-score on PolitiFact and by 7.21–21.74 in F1-score on GossipCop. Second, while model performance generally fluctuates depending on contexts and topics, our model maintains a consistent level of performance. Specifically, PolitiFact primarily focuses on fact-checking political news, whereas GossipCop covers entertainment and related domains. Many baseline models that perform well on one dataset fail to achieve similar results on the other. For instance, Hu et al.'s model, which ranks second to ours in F1-score on

PolitiFact, significantly underperforms on GossipCop, exhibiting a performance gap of 6.51 points in F1-score. In contrast, FENDI-2 shows only a 2.82-point difference between the two datasets, suggesting its stronger generalizability across different contexts. Altogether, we conclude that FENDI-2 effectively increases model performance.

4.4 Post hoc Analyses

To further assess the validity of our approach, we have conducted the following additional analyses: (i) A visual and distributional comparison between deceptive and non-deceptive fake news; (ii) an ablation analysis to evaluate the contribution of our newly proposed components to model performance; (iii) an application of our intent detection method, FENDI-1, to true news in order to verify whether it genuinely captures intent embedded in textual content than superficial cues; (iv) a sensitivity analysis of model performance under varying true news to fake news ratios; (v) evaluation of our model using additional datasets that capture the aforementioned challenges of concept drift, context dependency, and adversarial attacks; and (vi) a computational complexity analysis.

First, as an exploratory study to understand the characteristics of deceptive and non-deceptive fake news identified by FENDI-1, we utilize two tools: t-SNE and Kullback-Leibler (KL) divergence. The former is a dimensionality reduction technique for visualizing high-dimensional data, while the latter measures the difference between two probability distributions. Figure 5 presents the t-SNE plots for PolitiFact and GossipCop. In both Figure 5(a) and (b), we observe relatively clear separations between deceptive fake news instances (in red) and true news instances (in blue). However, many non-deceptive fake news instances (in green) appear on or near the boundary between deceptive fake news and true news instances. This supports our claim that non-deceptive fake news shares characteristics with both true and deceptive fake news, making fake news detection particularly challenging.

On top of that, Table 4 presents the KL divergence between different news classes. To measure KL divergence, we first quantify data instances belonging to various news classes based on their syntactic characteristics. Rather than relying on general embedding vectors generated by large LMs, which encode the mixture of semantic and syntactic information, we engineer exclusively syntactic features to validate our claim that non-deceptive fake news shares similar syntactic structures with true news, whereas deceptive fake news does not. For syntactic features, we employ eight commonly recognized indicators of deceptive intent from prior literature (Bachenko et al., 2008; Lee and Ram, 2021; Zhou et al., 2003). For each of the features, we compare the divergence between true news and non-deceptive fake news with that between true news and deceptive fake news. If the former shows a lower level of divergence than the latter, it suggests that non-deceptive fake news shares greater syntactic similarity with true news than deceptive fake news. According to the results, it shows that,

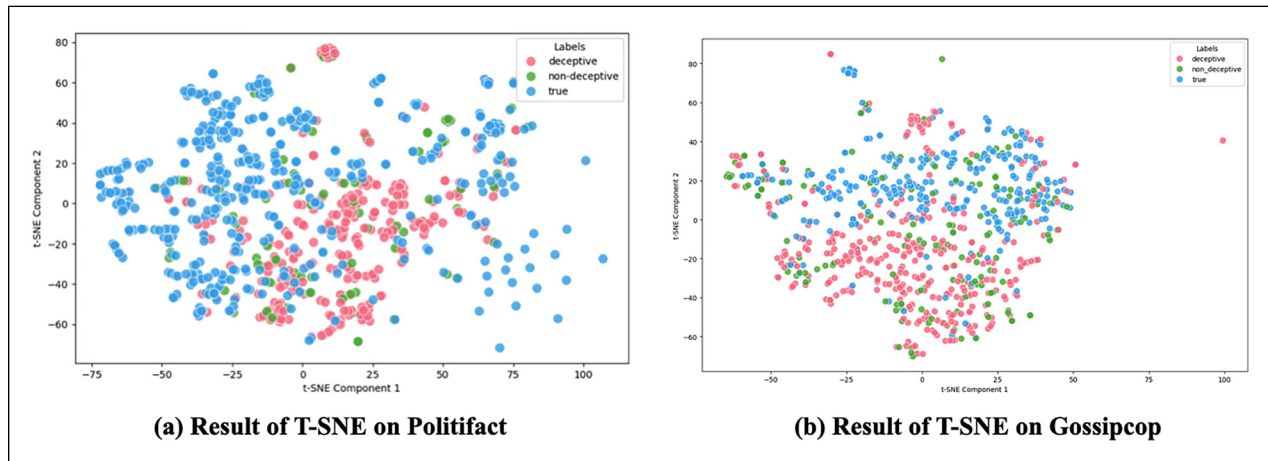
Table 3. Summary of results (fake news detection).

	Model	PolitiFact			GossipCop		
		Precision	Recall	F1	Precision	Recall	F1
Conventional ML	LSTM (Wang, 2017)	75.41	63.89	69.17	73.32	76.11	74.69
	CNN (Wang, 2017)	72.88	59.72	65.65	74.23	71.56	72.87
	Liu and Wu (2020)	75.00	70.83	72.86	75.76	73.68	74.71
Pretrain-and-Finetune	BERT	90.74	68.06	77.78	76.78	73.04	74.86
	DistilBERT	66.67	72.22	69.33	75.92	71.99	73.90
	Kaliyar et al. (2021)	72.22	77.78	74.90	73.70	77.34	75.48
Prompt learning	Peng et al. (2023)	79.15	58.40	67.21	72.16	52.76	60.95
	Hu et al. (2024)	92.73	70.83	80.31	74.68	72.94	73.80
	Lester et al. (2021)	91.84	62.50	74.38	77.60	66.28	71.49
FENDI-2		86.13	84.90	85.51***	83.72	81.68	82.69***

ML = machine learning; LSTM = long short-term memory; CNN = convolutional neural network.

*** indicates statistical significance at the level of 0.001. Note that we compare our model's F1-scores, obtained through five-fold cross-validation, with those of the second-best models for each dataset.

²The shaded areas indicate the highest scores for their corresponding performance measures.

**Figure 5.** t-SNE plots based on news classes.

in seven out of eight features, the divergence between true news and non-deceptive fake news has been lower than that between true news and deceptive fake news in both datasets. This further supports our methodological motivation that the syntactic heterogeneity of fake news increases the complexity of distinguishing true news from fake news.

Second, we performed an ablation analysis to evaluate the contribution of the components introduced in FENDI. Specifically, we examined the effect of removing (i) the referential DA component (i.e., FENDI-1) and (ii) the three-way classification component (i.e., FENDI-2) on model performance. Note that the latter experiment implies that neither FENDI-1 nor FENDI-2 is applied because FENDI-1 does not serve any functional purpose in the absence of FENDI-2.

To evaluate the contribution of FENDI-1, we implemented five alternative strategies for identifying the intent behind fake news. First, we applied a state-of-the-art unsupervised

learning technique, DeepCluster, which leverages backpropagation with pseudo labels to iteratively update model parameters (Caron et al., 2018). While the original study employed CNNs for image clustering, we adapted the architecture to use LSTMs to better suit the textual nature of our task. Second, we replaced FENDI-1 with the intent detection method proposed by Wang et al., which achieved the second-best performance in Table 2 for the review-to-news domain adaptation setting. Third, to rule out the possibility that performance gains were driven by random linguistic patterns or model behavior induced by ternary classification, we randomly assigned intent labels to fake news—with equal probabilities of being deceptive or non-deceptive—and then conducted model training and evaluation. Fourth, to account for the possibility that the performance improvement of FENDI-1 may be attributed solely to the inclusion of the contrastive learning component, we introduce an additional baseline in which contrastive

Table 4. KL-divergence between news classes.

	PolitiFact			GossipCop		
	T vs. NDF	T vs. DF	T-NDF < T-DF	T vs. NDF	T vs. DF	T-NDF < T-DF
Avg # of characters per word	0.34	0.61	Yes	0.21	0.18	No
Avg # of words per sentence	0.45	0.52	Yes	0.06	0.10	Yes
Avg # of clauses per sentence	0.46	0.35	No	0.09	0.22	Yes
Unique word ratio	0.50	0.60	Yes	0.09	0.12	Yes
Time indicator ratio	0.17	0.54	Yes	0.04	0.15	Yes
Self-referencing ratio	1.17	1.26	Yes	0.06	0.39	Yes
Third person referencing ratio	0.61	1.04	Yes	0.04	0.14	Yes
Past tense verb ratio	0.58	1.11	Yes	0.01	0.22	Yes

Table 5. Summary of ablation analysis.

		PolitiFact			GossipCop		
		Precision	Recall	F1	Precision	Recall	F1
FENDI		86.13	84.90	85.51	83.72	81.68	82.69
W/O FENDI-1	DeepCluster	78.16	83.69	80.83	75.08	76.25	75.66
	Wang et al.	84.21	82.05	83.12	80.05	76.25	78.11
	Random assignment	74.28	78.39	76.28	76.61	65.23	70.46
	W/O weight cloning	75.00	81.23	77.99	75.67	71.79	73.68
	W/O contrastive loss	82.58	83.76	83.17	79.43	79.96	79.69
W/O FENDI-1&2		81.60	75.78	78.58	79.59	67.89	73.27

learning is applied without the weight cloning layer. In this setting, f_{int_source} and f_{int_target} are identical, as the weight cloning process is omitted. Lastly, we test the contribution of the contrastive learning component to model performance by removing the contrastive loss from our weight cloning approach. In this case, the weights of f_{ref} and f_{int} will not be updated through contrastive loss (L_{rc}) but only through classification losses (L_{rsc} and L_{rtc}). On the other hand, to evaluate the effect of FENDI-2, we used the original binary dataset to train the same base model used in FENDI (i.e., ALBERT).

In Table 5, we present the results of our ablation analysis. The findings indicate that without the full set of components comprising our intent detection approach based on referential DA (i.e., FENDI-1), model performance drops significantly—from 2.34 to 9.23 points in F1-score for PolitiFact and from 3.00 to 12.23 points for GossipCop. That is, replacing FENDI-1 with alternative methods does not guarantee the same level of performance achieved with FENDI-1. Moreover, the performance drop when either the weight cloning or the contrastive loss component is removed suggests that optimal performance can only be attained when both components function in tandem. Furthermore, when the three-way classification component is removed and only the base model (i.e., ALBERT) is applied, the performance decline becomes even more pronounced. Specifically, the F1-score for PolitiFact decreases by 6.93 points, while that for GossipCop drops by 9.42 points.

Third, we applied FENDI-1, our intent detection model, to true news data to verify whether it genuinely captures

intent embedded in textual content rather than relying on superficial cues. While Table 2 already demonstrates that our approach outperforms other state-of-the-art intent detection methods—validated using both existing ground-truth data from the online review domain and newly generated ground-truth data in the fake news domain—these datasets have limitations. In particular, online reviews differ inherently from fake news in their linguistic and contextual structure, and the size of the labeled dataset for fake news remains small due to resource constraints.

Hence, to further validate FENDI-1's ability to identify genuine intent, we evaluated it using true news articles. This test rests on the assumption that the creators and disseminators of true news do not hold any intent to deceive others and thus exhibit non-deceptive intent. Therefore, if FENDI-1 accurately detects intent, it should consistently classify true news as non-deceptive.

We compare FENDI-1's performance against two strong baseline models—Wang et al.'s method and BERT—previously identified in Table 2 as the second and third best-performing models for intent detection, respectively. According to the result, FENDI-1 achieves an accuracy⁵ of 85.59% on the PolitiFact dataset, outperforming the baselines by 9.83 and 8.08 percentage points, respectively. On the GossipCop dataset, FENDI-1 achieves 87.07% accuracy, surpassing the baselines by 5.34 and 8.45 percentage points, respectively.

Table 6. Results of sensitivity analysis.

	Model	PolitiFact			GossipCop		
		Precision	Recall	F1	Precision	Recall	F1
2-to-1	Baseline	74.07	69.99	71.91	75.91	71.44	73.61
	FENDI	80.00	83.84	81.88	79.37	82.91	81.10
3-to-1	Baseline	65.00	68.42	66.67	73.31	74.23	73.77
	FENDI	71.11	84.21	77.11	77.85	82.73	80.22

These results offer two key implications. First, the fact that all three intent detection models—ours and the baselines—perform well above the random baseline (i.e., 50%) on true news data suggests that they are indeed capturing underlying intent rather than relying on incidental or confounding features. In particular, our model, FENDI-1, significantly outperforms the cutting-edge alternatives, reaffirming its superior capability in intent detection. Second, in comparison to the models’ performance on fake news (presented in Table 3), their performance on true news is noticeably stronger, highlighting the challenge of accurately detecting intent in fake news. This underscores the central role of intent detection in fake news classification and further validates the importance of our proposed approach.

Fourth, we performed a sensitivity analysis to investigate how variations in the distribution of true and fake news within the dataset affect model performance. Given that true news continues to constitute a larger proportion of news articles despite the rise of fake news, we evaluated our approach using the following true-to-fake news ratios: 2-to-1 and 3-to-1. In Table 6, we present a summary of the results. Note that, for brevity’s sake, we compare the performance of our model solely against the best-performing baseline model for each of the datasets, PolitiFact and GossipCop, which are Hu et al. (2024) and Kaliyar et al. (2021), respectively.

According to the results, we observe a performance decline for both the baseline and FENDI in PolitiFact, whereas performance remains relatively stable in GossipCop. Specifically, when the true-to-fake ratio was adjusted to 2-to-1 in PolitiFact, the baseline and FENDI experienced F1-score drops of 8.40 and 3.63 points, respectively. In contrast, the performance decline in GossipCop was far less severe, with decreases of only 1.87 and 1.59 points, respectively. A similar trend was observed in the 3-to-1 setting. We suspect that PolitiFact is more sensitive to class distribution changes than GossipCop due to its smaller dataset size, where even slight distributional shifts lead to substantial variations in results. However, even in PolitiFact, FENDI demonstrated relatively lower sensitivity to these modifications, exhibiting a smaller drop in F1-score.

Fifth, we further assess whether our model’s performance improvements stem from its ability to address key challenges in fake news detection—namely, concept drift, context dependency, and adversarial attacks—which we specifically targeted through the integration of the intent dimension. To this end, we

employ three distinct datasets, each designed to capture one of these challenges.

For concept drift, we evaluate our model using a rumor dataset developed by Zeng and Gao (2022), which includes 1198 rumor and non-rumor news articles verified on Snopes.com along with their associated tweets published over time. Some tweets were posted during the early emergence of rumors, while others appeared much later after the rumors became well established. This temporal variation provides an ideal setting to test the robustness of our model against evolving concepts over time (i.e., concept drift). We used the earlier half of the tweets for training and the later half for testing to examine whether a model trained on earlier rumors can generalize to later—and potentially drifted—versions of those rumors.

For context dependency, we utilize the dataset developed by Passaro et al. (2022), which consists of 568 news articles and tweets labeled as true or fake. What distinguishes this dataset is the inclusion of two additional annotation types—*in-context* and *out-of-context* labels. The *in-context* labels are assigned by annotators who possess sufficient background knowledge of the topics, while the *out-of-context* labels are assigned without such information. In some instances, all three labels (ground-truth, *in-context*, and *out-of-context*) align, whereas in others, discrepancies occur. Our focus is on the subset of topics where the ground-truth and *in-context* labels coincide but differ from the *out-of-context* labels. This distinction indicates that these topics are context-dependent and require additional background information to determine their veracity. From the full dataset, we identified 70 news articles and tweets (37 true and 33 fake) meeting this criterion, which were then used for our evaluation.

For adversarial attacks, we generated new datasets by manipulating the ones used in the main study (i.e., PolitiFact and GossipCop) through OpenAI’s GPT-5. Specifically, we employed the following prompt to modify the fake news articles in both datasets: “*Paraphrase the following article to conceal any deceptive intent without altering its content: [news article].*” The inclusion of these manipulated datasets allows us to extend the discussion related to Figure 2 and provides empirical support for our argument. It is important to note that, although real-world perpetrators may employ more sophisticated strategies to manipulate the content using various LMs, incorporating such techniques lies beyond the scope of this study.

Table 7. Results of experiments on model robustness against fake news challenges.

	Concept Drift (Zeng and Gao, 2022) ¹			Context Dependency (Passaro et al., 2022)			Adversarial Attacks					
							PolitiFact			GossipCop		
	Pr ²	Rc ³	F1	Pr	Rc	F1	Pr	Rc	F1	Pr	Rc	F1
Baseline	71.22	66.16	68.60	75.86	66.67	70.97	82.99	69.52	75.66	73.86	66.40	69.94
FENDI	79.63	74.10	76.76	78.13	75.76	76.92	86.23	82.05	84.09	83.33	81.96	82.64

¹Since the original dataset of Zeng and Gao (2022) includes over 3 million posts, we randomly selected 30,000 posts (approximately 1%) with an equal number of rumor (fake) and non-rumor (true) posts. Although this proportion appears small, we contend that, in absolute terms, a sample of 30,000 posts offers sufficient representativeness for our analysis.

²Precision.

³Recall.

Table 7 presents the results of our analysis. For brevity, we compare FENDI's performance only with that of Hu et al. (2024), which demonstrated the second-best performance in the main analysis. Overall, FENDI demonstrates greater robustness against the fake-news-related challenges compared to the baseline model, consistent with our expectations. In addressing concept drift, FENDI outperformed the baseline by 8.16 points in F1-score, indicating a higher level of performance stability when dealing with the same fake news topics evolving over time. Notably, FENDI's results are comparable to those reported by Zeng and Gao (2022; F1-score = 78.8), despite utilizing only 1% of the original dataset, further validating the effectiveness of our approach.

Against context dependency, our model outperformed the baseline by 5.95 points in F1-score, demonstrating its enhanced capability to detect fake news in context-dependent situations. Notably, although our model was tested only on the most challenging instances—those requiring contextual understanding for accurate identification as discussed above—it achieved even stronger performance than what has been reported in the original study, where models were evaluated on both simple and complex cases (F1-score = 76.50; Passaro et al., 2022). This further provides empirical support for the superiority of FENDI.

For the manipulated datasets simulating adversarial attacks, our model exhibited substantially stronger performance than the baseline. Specifically, on PolitiFact and GossipCop, FENDI outperformed the baseline by 8.43 and 12.70 points in F1-score, respectively. Although the manipulation led to a general decline in performance for both models compared with the main analysis (Table 3), the negative effect of the adversarial attacks was marginal for FENDI but considerably more severe for the baseline. This is evidenced by the larger F1-score gap between FENDI and the baseline when evaluated on the manipulated datasets relative to the original ones. Overall, the results reported in Table 7 provide compelling evidence of FENDI's robustness against key challenges in fake news detection.

Lastly, we have conducted a computational complexity analysis. For brevity, we do not report the full results here; however, our findings indicate that FENDI is computationally

efficient, requiring fewer resources compared to established large LMs. This further demonstrates the model's practical applicability. The detailed analysis is provided in Online Appendix D.

5 Conclusion

In this study, we proposed a novel methodological framework for fake news detection by addressing two interconnected challenges: The heterogeneous intent behind fake news and the lack of training data for intent detection. To tackle these issues, we introduced an intent-based fake news detection model that leverages a novel DA approach. Specifically, we incorporated the intent aspect by employing a two-step process: First, developing a DA model for intent detection, and using this model, transforming the fake news detection task from a binary classification problem (fake news vs. true news) into a ternary classification problem (deceptive fake news vs. non-deceptive fake news vs. true news). To address data scarcity in intent detection, we proposed referential DA, a novel method integrating contrastive learning, bootstrapping, and weight cloning to enhance intent inference through an effective DA strategy. By applying this approach, we could accurately identify the intent behind fake news even without sufficient training data, thereby improving overall fake news detection performance.

The implications of this study are manifold. From a methodological perspective, we contribute to the literature of fake news detection and DA. Specifically, by proposing the importance of intent in fake news detection and developing a novel DA method to effectively capture the intent, we have paved a new way for future studies in the field.

From a theoretical standpoint, our study advances the prior studies on false information and intent by extending discussions on fake news heterogeneity through the lens of intent. This encourages the integration of social science perspectives into computational approaches for identifying and analyzing fake news.

From a managerial perspective, beyond its significance in mitigating fake news that disrupts business operations, this

study underscores the importance of caution when implementing automated fake news moderation. Recently, social media platforms, in collaboration with political authorities (e.g., the European Commission), have focused on improving fake news analysis, automated detection, and content removal (European Commission, 2020). However, such measures remain controversial due to their implications for freedom of speech. Based on our findings, business managers and policymakers should critically reconsider key questions such as: *Can fake news regulation be justified without accounting for deceptive intent?* and *If no deceptive intent is involved, should individuals be allowed to share information they believe to be true, even if it is false?* These considerations highlight the need for a more nuanced approach to addressing fake news in policy and business strategies.

This study, however, is not without limitations, which open avenues for future research. One such limitation concerns the applicability of our DA approach, which depends on the existence of a suitable FoR—defined as a field whose task differs from that of the FoI but shares the same source and target domains. Additionally, the criteria for selecting an appropriate FoR when multiple candidates are available have not been thoroughly addressed. In this study, the choice of FoR was based on data availability. However, when several options exist, identifying the task characteristics that make a FoR more effective for knowledge transfer—such as similarity in task complexity—remains an important question. While this exploration lies beyond the scope of the present study, which primarily aims to introduce and evaluate the concept of FoR and weight cloning in the context of DA, it warrants careful examination in future work.

In addition, temporal gap and error propagation introduced during intent classification (i.e., Step 1) may influence and confound the interpretation of the fake news classification results. Specifically, the dataset used to train FENDI-1 for intent classification originates from data collected a decade ago, creating a temporal mismatch between training and inference. Although we posit that intent classification primarily relies on relatively stable textual characteristics, such as syntactic features, they could nonetheless evolve over time, potentially attenuating performance gains. Also, misclassification of intent, which can be caused by various factors, including the temporal gap issue, may propagate to the downstream task of fake news classification, thereby degrading overall model performance. While we have conducted multiple post-hoc analyses to assess the robustness of our approach to these concerns, it remains important to consider the issues in practice.

Furthermore, the interpretability and explainability of FENDI could be further improved (Lee and Ram, 2024). Although we validated our model through extensive experiments and empirical results, we did not investigate in detail the internal mechanisms—particularly the neural network dynamics—that underlie the model's effectiveness in capturing the intent dimension of fake news. While developing an explainable ML framework to uncover such mechanisms lies beyond

the scope of this study, future research could extend our work by examining the inner workings of FENDI to provide deeper insights into its decision processes.

Lastly, while our approach is particularly well suited for contexts in which individuals deliberately consume and propagate fake news, its practical effectiveness in settings characterized by heuristic consumption—i.e., when individuals engage with fake news without deliberate or systematic thought—remains uncertain. In such cases, individuals are less likely to convey their intent explicitly in their writing (Pennebaker, 2016), which may limit the utility of intent-based detection methods. Although prior studies have indicated that fake news consumption and dissemination are often grounded in systematic cognitive processes rather than purely heuristic behavior (Giglietto et al., 2019), it remains an open and important question whether approaches that rely on detecting the intent of fake news propagators can still perform effectively in scenarios where intent is ambiguous or concealed.

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Notes

1. We refer to conventional ML as methods that require a customize neural-network architecture to be trained from scratch on task-specific data.
2. The data instances shown in the figure were generated using Prolific, a crowdsourcing platform for researchers, ensuring ground-truth labels for both veracity and intent. A more detailed explanation of the data collection process will be provided in a later section.
3. The text was produced using the following prompt: "Paraphrase the following article disguising any deceptive intent while not changing the content: [news article]."
4. Although this score is substantially higher than that of the baselines, we sought to assess whether it represents an acceptable level of performance. To this end, we recruited three human workers (graduate students in information systems) to independently identify the intent of each data instance, aggregating their judgments through majority voting. The results showed that the human workers achieved precision, recall, and F1-scores of 73.33, 39.29, and 51.16, respectively, indicating a tendency to classify texts as having non-deceptive intent (i.e., low recall). These findings

underscore the inherent complexity of identifying intent in text and further demonstrate the effectiveness of FENDI-1.

5. It is important to note that in this context, accuracy is equivalent to recall, while the precision is always 1. This is due to the binary classification setup, where non-deceptive true news is labeled as the positive class and deceptive true news as the negative class, yet no actual negative samples are present. As a result, we report accuracy as the sole performance metric in this analysis.

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